

SAI NAVEENA KODE

📞 631-739-4572 ✉️ sainaveena.kode@stonybrook.edu 🔗 [linkedin.com/in/naveena-kode/](https://www.linkedin.com/in/naveena-kode/)

Education

Stony Brook University, NY

August 2024 – May 2026

Master of Science in Data Science

GPA: 3.48

Relevant Coursework: NLP, Big Data Systems, Data Analysis, Statistical Computing, Distributed Systems, Data management

Jawaharlal Nehru Technological University Hyderabad, India

August 2020 – July 2024

Bachelor of Science in Computer Science

GPA: 3.3

Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Database Management Systems, Operating Systems, Artificial Intelligence, Computer Networks, Cloud Computing, Machine Learning, Computer Organization

Experience

Stony Brook University - Digital Services

August 2025 – Present

Research Assistant

Stony Brook, NY

- Designed and deployed a Retrieval-Augmented Generation (RAG) system that improved search and discovery efficiency by 20% across digital collections.
- Engineered pipelines for document preprocessing, vectorization, and semantic retrieval, achieving 85% relevance accuracy in context-aware responses.
- Collaborated on integrating the prototype into multiple digital services workflows, preparing for scalability to thousands of archival records.

Stony Brook University

August 2025 – Present

Research Assistant

Stony Brook, NY

- Engineered prototype backend architecture using SQL and MongoDB to structure user profiles, MCQ responses, and learning analytics, enabling data-driven personalization.
- Created wireframes and interactive UI mockups with Figma and Streamlit, establishing user flows and accessibility features for the intelligent tutoring platform.
- Built initial Retrieval-Augmented Generation (RAG) pipeline leveraging Python, LangChain, and OpenAI APIs, laying the foundation for adaptive tutoring and real-time feedback.

Meta Business Solutions

June 2022 – October 2022

Intern

Telangana, Hyderabad

- Developed and implemented prototype Data Management Plans using SQL and Excel-based validation frameworks, ensuring accuracy and improving overall data quality by 20%.
- Led data management initiatives by designing workflows in Python (Pandas, NumPy) optimizing consistency and efficiency, resulting in a 25% increase in processing speed.
- Established automated data validation scripts and dashboards with Python, SQL, and Power BI, reducing entry errors by 30% and enhancing reliability of organizational datasets.

Projects

Context-Preserving Document Layout Extraction for LLM Consumption: | *Python, PaddleOCR, PPStructureV3*

- Devised a Streamlit web application to extract structured content from PDFs, scanned images, and Office documents using PaddleOCR and PPStructureV3.
- Established advanced table parsing and export workflows with BeautifulSoup and Pandas, generating downloadable JSON outputs for seamless LLM integration.

Fault-Tolerant Distributed Transaction System | *Python, SQLite3, gRPC, Paxos*

- Assembled a distributed transaction system implementing Paxos and Two-Phase Commit (2PC) protocols with concurrency control and failure recovery using Python, SQLite3, and gRPC.
- Integrated cross-shard communication and coordination logic to ensure fault tolerance, consistency, and reliability in a partitioned banking environment.

Decentralized Subscription Service | *React.js, Solidity, Web3.js, Node.js*

- Built a React-based web app with a media player and subscription workflow, integrating Web3.js, MetaMask, and Solidity smart contracts for blockchain-based payments.
- Deployed content streaming with HTML5 video player, Node.js APIs, and IPFS/CDN storage, enabling a simple end-to-end decentralized subscription experience.

Technical Skills

Programming & ML: Python (NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV, LangChain), R, Java, C++, JavaScript, SQL; Deep Learning (CNNs, MediaPipe), NLP, RAG, Distributed Systems (gRPC, Paxos, 2PC)

Data & Databases: Data Analysis Visualization (Excel, Streamlit, Power BI, Matplotlib, Seaborn, EDA); Databases (MySQL, MongoDB, SQLite3)

Tools Systems: Operating Systems, Computer Networks, Blockchain (Solidity, Web3.js, MetaMask), Frontend (React.js, HTML5, CSS/Tailwind), Backend (Node.js, Express), Cloud Storage (IPFS/CDN)